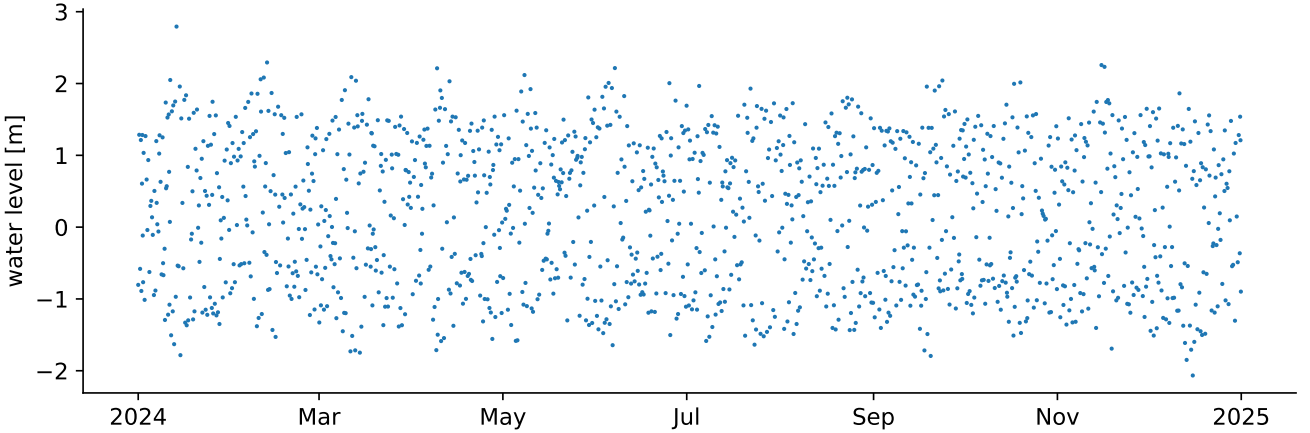
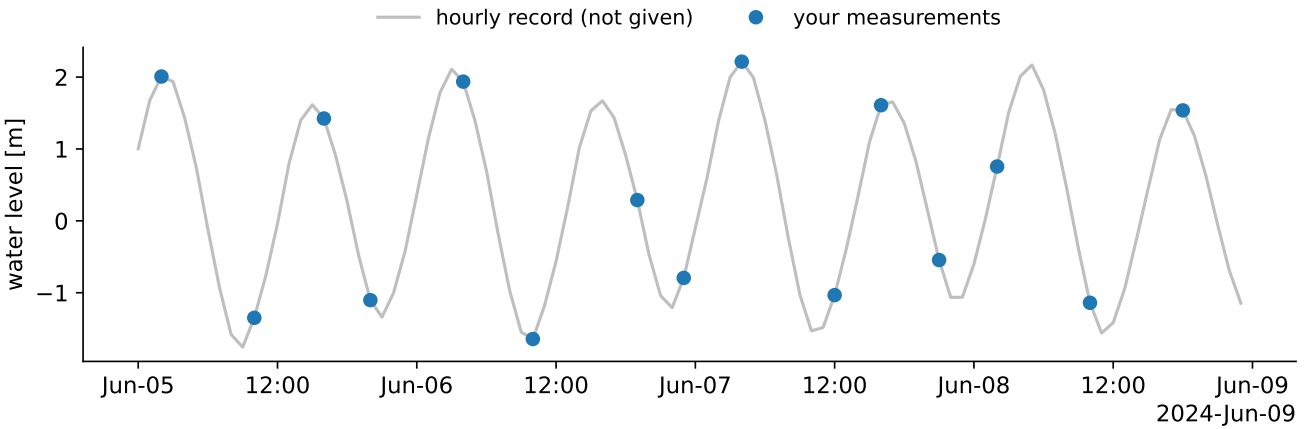
Foundations of Data Science

Case study: Tides, storm surge and sea level

LIÈGE université



The case study: one year of sea level in the harbour of Boston, 1443 measurements taken at random times, 3 to 9 hours apart.

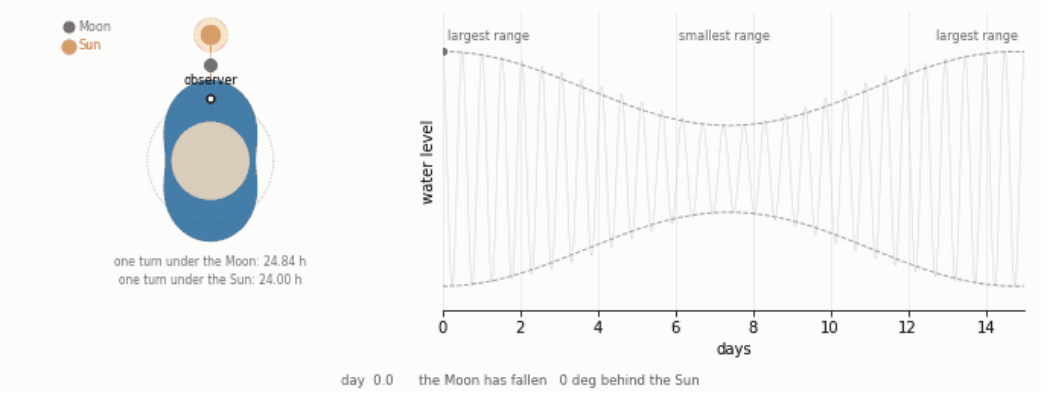


Four days in June. The sea rises and falls by about 3 m twice a day, but each tide is caught by one or two measurements only.

What moves the sea

- **The tide.** The Moon and the Sun raise the sea twice a day. The tide is a sum of sine waves, the **constituents**, whose periods are known from astronomy, for example 12 h 25 min for M2 (Moon) and 12 h for S2 (Sun). Their size and timing must be learned from the data.
- **The surge.** The weather pushes the sea up, by about 1 cm for each hPa of pressure drop, plus the push of the wind. It builds up in a few hours, fades over a day or two, and never repeats.
- **The slow level.** The average sea level changes over the year, and it has risen since the zero of the station was set.
- **The fast movements.** Waves and water sloshing in the harbour look like noise.

> The gauge measures only the sum. Nobody measures the tide or the surge alone.



M2 and S2 go in and out of step every 14.77 days:
large tides, then small tides.

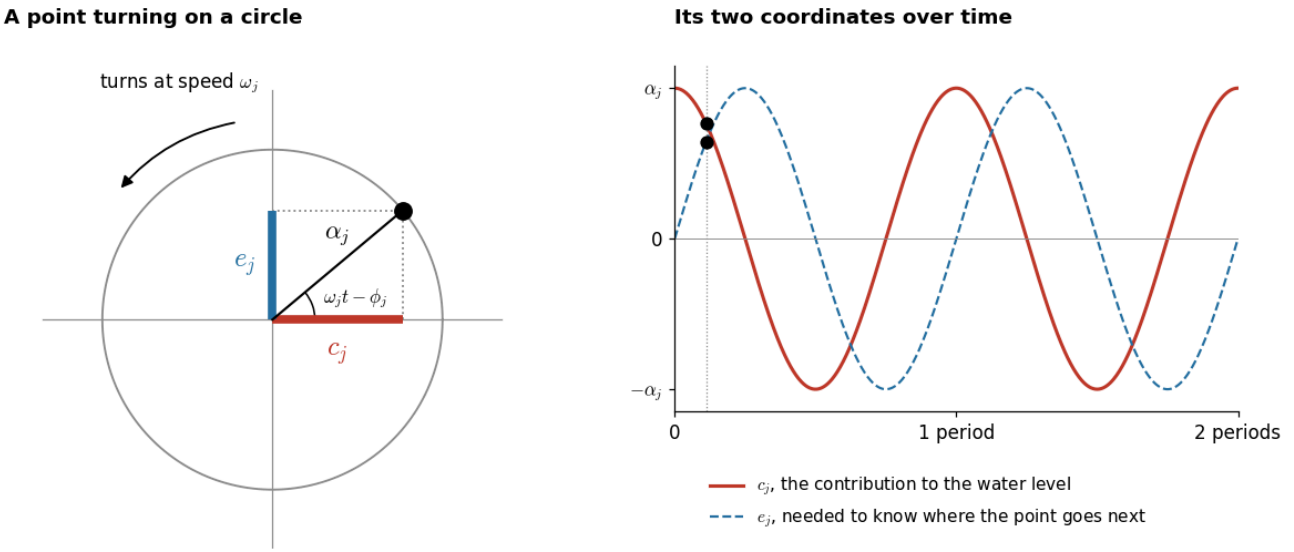
Why this is a data science problem

Flood warnings need the surge, but the surge is never measured. It is hidden in a sum, together with the tide and the slow level.

The tide, the surge and the slow level are **latent states** that change over time, and each measurement is their sum plus noise.

> This is a state-space model, as in Lecture 5. The Kalman filter and smoother separate the components, and say how sure they are about each one.

The model



Each constituent is a point turning on a circle. Its horizontal coordinate c_j is its share of the water level. The state stacks all the components,

$$\mathbf{z}_t = (c_{1,t}, e_{1,t}, \ldots, c_{J,t}, e_{J,t}, s_t, \ell_t),$$

with the surge s_t and the mean level ℓ_t.

Between two measurements, Δt hours apart, each point turns by $\omega_j \Delta t$ and the surge relaxes towards zero, as the wolf of Lecture 5 returns home. Both are solved exactly,

$$p(\mathbf{z}_t | \mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{z}_t | \mathbf{A}\mathbf{z}_{t-1}, \mathbf{Q}), \quad \text{with } \mathbf{A} \text{ and } \mathbf{Q} \text{ that depend on } \Delta t.$$

Each measurement adds the components, the effect of the air pressure and a noise,

$$x_t | \mathbf{z}_t \sim \mathcal{N}(\mathbf{H}\mathbf{z}_t + \beta \tilde{p}_t, \sigma_x^2).$$

The parameters $\theta = (\kappa, \sigma_s, \sigma_{\text{tide}}, \sigma_x, \beta)$ are estimated from the data.

What you will do

Build

- Q1. Explore the data
- Q2. Choose the constituents
- Q3. Write the model, and build $\mathbf{A}$, $\mathbf{Q}$, $\mathbf{H}$
- Q4. Choose a prior, and simulate

Compute

- Q5. Kalman filter
- Q6. Smoother, tested on simulated data
- Q7. Estimate θ
- Q8. Separate tide, surge and slow level

Critique

- Q9. Check the model against the data
- Q10. Forecast the water level 48 h ahead

Practical

- **Deadline.** December 18, 2026, on Gradescope.
- **Notebook.** `project/case-study.ipynb`. Each point has a title cell, such as `### QUESTION 3.A`, and your answer goes below it. The notebook must run from top to bottom, and keep its outputs.
- **Code.** `numpy`, `jax` or `pytorch`, as you prefer. Templates of the functions are given, and you may change them.
- **AI assistants** are allowed. You must be able to explain every line and every choice.
- **Evaluation.** A 30-minute oral discussion in January, separate from the theory exam. The case study counts for 40% of the final grade, and needs at least 8/20.

Tips

- Start from the wolf. Lecture 5 and `nb05` contain the filter, the smoother and the exact discretization that you need.
- Test the filter on simulated data (Q6) before you use it on the real record.
- Write the slow cells, the estimation and the simulations, so that you can run them again alone.
- Start early. Everything after Q5 depends on the filter.

Questions?